Evan Bell

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Education

Massachusetts Institute of Technology May 2025
S.B., Computer Science and Engineering; Mathematics (double major) *Cambridge, MA*
GPA: 4.8/5.0

Research Experience

MIT Kellis Lab – MANTIS *Spring 2024*

- Implemented recursive K-means clustering and model-generated hierarchical labels for an interactive document map in Python and TypeScript, using cluster contents and parent-category context.
- Optimized Three.js rendering of thousands of interactive points using InstancedMesh, cached cluster boundaries, viewport culling, and zoom-dependent detail.

MIT Deng Energy and Nanotechnology Group *2022-2023*

- Developed a Python pipeline converting parametric structure specifications into G-code; fabricated samples for combustion research and verified geometry using measurements and microscopy.
- Designed CAD fixtures and 3D-printed mounts for a motorized syringe printhead, integrating existing components into an experimental fabrication setup.

Engineering Experience

Google – YouTube Aug 2025 – Present
Software Engineer *San Bruno, CA*

- Designed and implemented C++ backend ranking and selection components, integrating revenue signals and rule-based prioritization through production RPC interfaces.
- Built an offline reconciliation pipeline using Flume and Spanner to detect and repair inconsistencies in user data records.

SESCO Enterprises Summer 2025
Software Engineering / Quantitative Research Intern

- Profiled Python/Numba cycle-space power-flow software; implemented experimental optimizations and compared numerical error with reference solvers on fixed grid cases.

Dimensional Insight Summer 2024
Software Engineering Intern

- Selected SQLite and implemented schema, legacy-file migration, and C integration for user/group configuration storage; observed startup fall from roughly five minutes to ten seconds on the same machine.

Selected Algorithmic Projects

Quoridor – Search and Self-Play Agent *Python, PyTorch, NumPy, Numba*

- Independently implemented MCTS, policy/value networks, and self-play training; experimented with CNN, residual, and transformer architectures and alternative board encodings.
- Built an engine reaching approximately 50,000 player actions/s on a MacBook, including legality updates and win detection, using cached wall legality and Numba-compiled DFS.

MIT PokerBots – Two-Person Team *IAP 2022*

- Implemented CFR from scratch and preflop hand abstractions using simulated outcome distributions and Earth Mover's Distance to reduce the strategy space for training.

Technical Skills

Languages: C++, Python, C, TypeScript, SQL
Libraries / Systems: PyTorch, NumPy, Numba, Three.js, SQLite, Flume, Spanner, RPCs, Git, Linux
Activities: MIT Division I Varsity Heavyweight Crew; MIT Poker Club